FUROSEMIDE

Monitor potassium

Ototoxic if fast IV

CHEMICAL NAME 4-chloro-2-[(furan-2-ylmethyl)amino]-5-sulfamoylbenzoic acid
CLASS Loop diuretic; also called frusemide
TRADE NAMES Lasix · Furosemide (generic) · Frusol (oral solution)

THERAPEUTIC USES / INDICATIONS

- Acute pulmonary oedema and congestive heart failure.
- Oedema from liver cirrhosis, nephrotic syndrome or renal disease.
- Oliguria in renal failure (to promote diuresis).
- Hypertension (resistant); hypercalcaemia (with fluids).

ADMINISTRATION

- **PO** in the morning (second dose by early afternoon).
- **IV** bolus slowly: max 4 mg/min for doses >80 mg (ototoxicity).
- **IM** possible; protect from light; do not use discoloured solution.

DOSAGES

- **Oedema PO:** 20–40 mg once daily; increase as needed (80 mg+ in resistant cases).
- **Acute pulmonary oedema IV:** 40 mg slowly over 1–2 min, repeat or increase per response.
- **Infusion:** 5–40 mg/h (resistant oedema).
- **Children:** 0.5–1 mg/kg per dose every 6–12 h (max 6 mg/kg/day PO).
- Elderly: start low.

PHARMACOKINETICS

- **Onset:** IV 5 min; PO 30–60 min. **Peak:** IV 30 min; PO 1–2 h. **Duration:** ~6 h ("Lasix lasts six").
- Highly protein-bound; partly metabolised in kidney.
- Excreted in urine. **t½:** 1–2 h (longer in renal failure).

PHARMACODYNAMICS

- Inhibits the **$Na^+/K^+/2Cl^-$ co-transporter** in the thick ascending loop of Henle → excretion of up to 25% of filtered sodium; water follows.
- Venodilation (↓ preload) before diuresis in pulmonary oedema.
- ↑ excretion of K^+, Mg^{2+}, Ca^{2+}.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Frequent urination, thirst.
- Dizziness, postural hypotension.
- Muscle cramps.

ADVERSE EFFECTS (SERIOUS)

- Hypokalaemia, hyponatraemia, hypomagnesaemia, metabolic alkalosis.
- Dehydration, AKI (pre-renal).
- Ototoxicity (tinnitus, deafness) with rapid IV or high doses.
- Hyperuricaemia (gout), hyperglycaemia; pancreatitis, photosensitivity (rare).

CONTRAINDICATIONS & CAUTIONS

- Anuria; severe hypovolaemia or dehydration.
- Severe hypokalaemia or hyponatraemia.
- Hepatic coma/pre-coma.
- **Caution:** sulfonamide allergy, gout, diabetes, prostatic hypertrophy (retention).

KEY INTERACTIONS

- **Digoxin:** hypokalaemia → digoxin toxicity.
- **Aminoglycosides, vancomycin:** ↑ ototoxicity/nephrotoxicity.
- **Lithium:** ↑ levels.
- **NSAIDs:** ↓ diuretic effect; **steroids, salbutamol:** ↑ K^+ loss.

NURSING MANAGEMENT

- Check K^+ and BP before giving; hold and report if K^+ <3.5 mmol/L (or per policy).
- Daily weight (same time, same scale) and strict fluid balance.
- Monitor U&E, glucose, uric acid.
- Pulmonary oedema: monitor RR, SpO_2, breath sounds, urine output (catheter may be needed).
- Assist with mobility (postural drop); ensure toilet access.

PATIENT EDUCATION

- Take in the morning to avoid night-time urination.
- Weigh daily; report gain >1 kg/day or 2 kg/week.
- Eat potassium-rich foods unless told otherwise.
- Report cramps, weakness, palpitations, ringing in ears.

KEY POINTS

- Loop diuretic; blocks NKCC2 in loop of Henle.
- Hypokalaemia (+ digoxin = toxicity).
- IV onset 5 min; max 4 mg/min.
- Daily weight and fluid balance.

SOURCES CHECKED [325] Furosemide 10 mg/ml Solution for Injection (UK SmPC) [326] Furosemide, IM, IV (US label) [327] Furosemide, oral (US label)

Full references p. 432–457